Physical-Event Contracts Before Policy Learning: A Mechanical Timing and Recovery Component Study

Research development draft [1]

Abstract

Laboratory deadlines concern physical effects, not just commands. Eight native mechanical development witnesses satisfy independently scored original-event contracts using one strong public rule batch; producer and independent reintegrations reproduce state and time exactly. Legal public-model inference and blind redundancy explain these cases without queries or learned weights. We preserve qualification, reachability and attribution failures. These diagnostics establish scoped reachability, not VLA learning, chemical completion or unseen-task generalization.

1. Problem and positioning

Command receipt, physical termination and chemical completion have different times. MATTERIX already supplies chemistry twins, waiting, concurrent workflows and process verification, and separates heater-off from reaction completion (Darvish et al., 2026). Chronos and Time as a Control Dimension already investigate physical time and punctual control (Chronos, 2026; Time Control, 2026). We ask whether verified recovery experience improves a shared policy beyond matched-information control and matched-budget data. This component report does not evaluate that learning claim.

2. Contracts and causal information

A task declares predicates, graph, observations and resources. Occurrence intervals $I_e = [\ell_e, u_e]$ share one epoch. For target D,

$$I_{\text{err}} = [\ell_b - u_a - D,\ u_b - \ell_a - D].$$

Containment in $[-\epsilon, \epsilon]$ is TRUE; disjointness is FALSE; partial overlap is UNKNOWN. Missing/invalid events cannot succeed. Joint scoring requires original references and physical/graph predicates; refusal, UNKNOWN and exhaustion remain in the episode denominator.

Our CPU MuJoCo fixture contains three contact-free motors. B's first actual $q = 2$ crossing is the original event; nonnegative B torque and no state/time rewrite make it irreversible under the interface. A/C torque changes only at ACK; the first A transaction fails physically. Delayed public packets contain position, velocity, torque, timestamps and reservations. ACK has no effect-success flag; referee RAW is private. Public B $[.40, .45]$ s arrives at .55 s. A predeclared nominal model conditionally estimates the *same* event as $[.44, .45]$ from legal packets; the coarse record persists. Every baseline may use this model. Packet compatibility is not model-support certification.

For C's $D \pm \epsilon$, A's upper bound U_A, C-to-A gap g and reference/C/A box widths w, c, a, combining lower/upper bounds yields

$$w + c + g + a \le U_A - D + \epsilon.$$

This necessary box condition is not resource feasibility. Two possible A effects from 50 ms-separated ACKs initially have a roughly 60 ms union; public torque feedback can select a 10 ms box. Coarse-box refusal is neither physical impossibility nor an information lower bound.

3. Diagnostics and three failure classes

The v2 diagnostic independently crosses graph order, B service 50/450 ms and $D = 2.8/3.1$ s: eight fixed configurations, not random seeds. A must stop $[1.8, D]$ after B, C within $D \pm .15$, with a 40 ms edge. B's reservation starts at .65 s; declared torque and motion persist through its event plus six seconds. One public batch allows two A requests for both none/one-drop hypotheses; redundancy is not diagnosis.

All eight witnesses are joint TRUE: four per graph. Each 12 s trace has 1,201 RAW samples. Sixteen producer/third-party command replays match state/time exactly: 24 epochs, 28,824 rows; primary remains eight. Source was frozen first. Stops require prior movement and 21 low-speed samples spanning 200 ms. Onset and confirmation differ; sampled

[1] Internal component report; chemical and learning validation remain open. Correspondence to: Author details pending.

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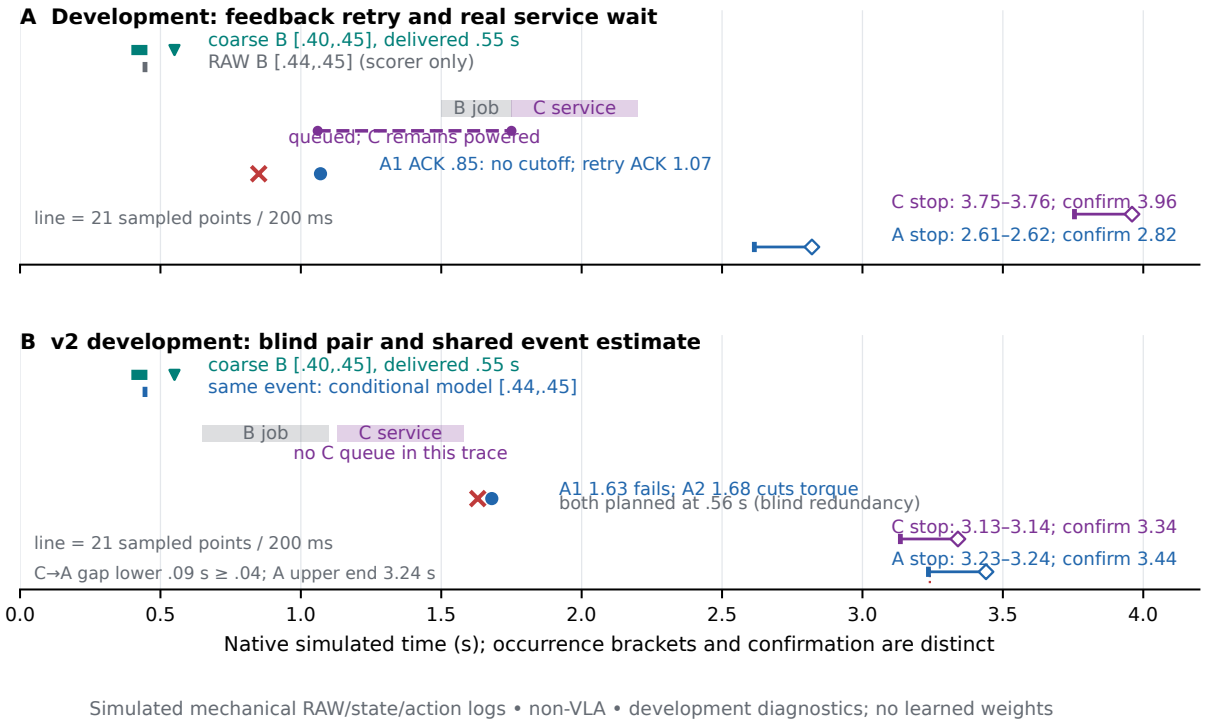


Figure 1. Recorded development timelines: 4.2 s of complete 12 s traces. A: ineffective ACK, feedback retry and powered C queue. B: same-event inference and preplanned redundancy. Short marks: occurrence brackets; open diamonds: 21-point/200 ms confirmation. Simulated mechanical state/action; no RGB, VLA, hardware or learned-policy claim.

maintenance does not guarantee intervals between samples. Each witness charges two decision ticks and 282–370 proposals; host computation otherwise pauses the plant.

Qualification. A prior rigid-bottle qualification passes mechanical criteria but has three EARLY timing scores and FALSE strict sampled hold. Supported stable release onset cannot become first unloading or a contact fraction.

Reachability. The old resource draft retains six unsupported conditions out of eight. A .75 s candidate still has a high-busy C-to-A counterexample: post-slot C ends ≥ 3.21 s, but the deadline/edge require ≤ 3.19 s; pre-slot is too early. These drafts are not native trials; .65 s has native witnesses.

Attribution. Public-model inference removes the coarse-reference restriction here. One rule batch solves all eight without queries or training: no agentic necessity or advantage, and no demonstrated post-failure adaptation.

4. What a learning claim still needs

Seen nominal configurations are development data; repeating 32 trials would compare mechanisms, not unseen tasks. Preserve any null advantage over strong event/HSM control. Freeze observer/model/harness/gains; separate weight updates from retained/reset context. Match backbone/data, labels, updates, interactions and compute across DP/history, offline imitation, DAgger and student-state learning. Aggregation alone is not distinct OPD. Independently calibrate flow, loaded RPM, transfer and chemical endpoints; these rotors do not certify C5/C6/C13. Formal32 trials, fit, GPU, LLM and weight updates here: zero. Independent review/replay pass; the audit is post hoc, not prospective.

References

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Chronos Policy: Learning Robot Policies on Physical Time. arXiv:2610.13929, 2026.

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